Week 8 – Mid-Course Review & Simple Algorithms

■ Learning Objectives

• Review foundational programming concepts (Weeks 1–7).
• Understand brute-force and counting-based algorithms.
• Learn efficiency basics and debugging strategies.
• Apply problem-solving to CCC-style questions.

■ Review Summary (Weeks 1–7)

• Variables, Data Types, Input/Output.
• Conditionals and Boolean Logic.
• Loops (for, while, nested).
• Strings, Lists, Dictionaries.
• Functions and 2D Lists.

Example 1 – Brute Force Search

```
# Find two numbers that sum to target
nums = [2, 7, 11, 15]
target = 9
for i in range(len(nums)):
    for j in range(i+1, len(nums)):
        if nums[i] + nums[j] == target:
            print(nums[i], nums[j])
```

Example 2 – Counting Occurrences

```
text = "banana"
count = 0
for ch in text:
    if ch == 'a':
        count += 1
print(count)
```

■ Practice Problems

Practice 1 – Sum of Multiples

Find the sum of all numbers under 100 that are multiples of 3 or 5.

```
total = 0
for i in range(1, 100):
    if i % 3 == 0 or i % 5 == 0:
        total += i
print(total)
```

Practice 2 – Count Words

Read a sentence and count the number of words.

```
sentence = input("Enter sentence: ")
words = sentence.split()
print("Word count:", len(words))
```

Practice 3 – Debug the Code

Find and fix the bug that causes incorrect total output.

```
nums = [1, 2, 3, 4]
total = 1
for n in nums:
    total = total + n
print("Sum =", total)  # Should be 10, not 11
```

■ Quiz – 10 Questions

1. What function is used to take user input?
2. Which operator checks for equality?
3. Name two looping structures in Python.
4. What does range(5) return?
5. How do you access the last element of a list named nums?
6. What method converts a string to lowercase?
7. What keyword exits a loop immediately?
8. Define a function that doubles a number.
9. What does len([1,2,3,4]) return?
10. What is brute force algorithm in one line?

■ *Answer Key*

1. input()
2. ==
3. for, while
4. 0,1,2,3,4
5. nums[-1]
6. lower()
7. break
8. def double(x): return x*2
9. 4
10. Tries all possible solutions until one works.

■ Homework Challenge

Write a program that counts how many numbers between 1 and 1000 are divisible by both 3 and 7.

```
count = 0
for i in range(1, 1001):
    if i % 3 == 0 and i % 7 == 0:
        count += 1
print("Count:", count)
```